

SemEval-2027 Task Proposal: DiCo-NLI—Directional Consistency in Fine-Grained Natural Language Inference

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1 Overview

Task. DiCo-NLI evaluates direction-sensitive fine-grained natural language inference (NLI) over ordered phrase pairs. For each source pair (A, B) , systems predict a label for both the original order (A, B) and the reversed order (B, A) . The reversible label algebra is deterministic: EQUIV maps to itself, while FORWARD and BACKWARD map to each other. The shared task therefore measures both item-level label correctness and whether predictions remain compatible under reversal.

Motivation. NLI systems and large language models can obtain strong aggregate scores while relying on shallow cues, annotation artifacts, or incomplete semantic representations (Gururangan et al., 2018; Lopez-Gazpio, 2024). Direction is a central part of NLI meaning: if *everyone is hungry* entails *someone is hungry*, the reversed ordered pair does not have the same label. DiCo-NLI targets this direction-sensitive failure mode in a compact, auditable, and multilingual setting.

The task is inspired by reversal-based evaluation, including the Reversal Curse of Berglund et al. (2024), but it is not a direct benchmark of parametric factual reversal. Berglund et al. evaluate whether a model trained on a fact such as “A is B” can retrieve or generate the reverse form “B is A” from its parameters. DiCo-NLI instead supplies both phrases and asks for their ordered inferential relation. The connection is methodological rather than identical: reversal is used as a controlled stress test for directional generalization. Follow-up work has studied reversal failures and mitigation strategies in factual recall and question-answering settings (Golovneva et al., 2024; Lin et al., 2024). DiCo-NLI complements that line by operationalizing reversal in supervised fine-grained NLI.

Revision focus. This revised proposal addresses the conditional-acceptance reviews by sharpen-

ing the conceptual scope, renaming the task from CoCo-NLI to DiCo-NLI, replacing “coherence” with the more precise term “consistency”, making the metric roles explicit, adding pilot results, specifying contamination controls, defining multilingual tracks, and detailing translation validation and baseline resources. The official task target is directional consistency in fine-grained NLI, not a claim that DiCo-NLI scores are interchangeable with Berglund-style parametric Reversal Curse scores.

Expected participants and impact. The task should attract researchers working on NLI, semantic evaluation, multilingual NLP, interpretability, model robustness, and LLM reasoning diagnostics. It contributes a reusable benchmark where label correctness and reversal consistency can be analyzed separately, with English, Spanish, Basque, and mixed-language tracks, an official scorer, validation tools, and runnable starter baselines. The task is designed to remain useful after SemEval as a diagnostic resource for directional generalization and multilingual fine-grained inference.

2 Relation to Prior Work

Fine-grained NLI and PhrasIS. DiCo-NLI builds on Natural Logic-style fine-grained inference labels (MacCartney, 2009) and on PhrasIS, a phrase inference and similarity benchmark derived from naturally occurring image captions and news headlines (Lopez-Gazpio et al., 2024). PhrasIS re-annotates phrase pairs from the Interpretable STS lineage (Lopez-Gazpio et al., 2017), where sentence pairs are decomposed into aligned semantic chunks such as nominal expressions, verb chains, prepositional phrases, adverbial expressions, and quantificational expressions. DiCo-NLI is compositional in this phrase-level sense: it evaluates inference over local semantic units built from smaller lexical and syntactic material. It is not intended as

Label	Source	Ordered	Rev($\cdot$)
EQUIVALENCE	686	1,372	EQUIVALENCE
FORWARD	624	1,260	BACKWARD
BACKWARD	636	1,260	FORWARD
Total	1,946	3,892	–

Table 1: English reversible source data and label-reversal algebra. “Ordered” counts both directions.

a controlled benchmark of arbitrary compositional generalization over generated templates.

Consistency-based NLI evaluation. DiCo-NLI is also related to consistency evaluation for language models. BECEL (Jang et al., 2022) evaluates several behavioral consistency properties, including symmetric consistency for standard three-way NLI. DiCo-NLI differs by explicitly modeling direction-sensitive label reversal in a fine-grained label space: FORWARD_ENTAILMENT and BACKWARD_ENTAILMENT swap under reversal, while EQUIVALENCE is invariant. Recent work on NLI meta-inferential properties (Blanck et al., 2026) and transitive self-consistency (Wu and Last, 2025) is complementary, but remains in the research-benchmark setting. DiCo-NLI brings a multilingual, fine-grained, directional-reversal setup into a shared-task format with official data, scorer, validation tools, and participant submissions.

3 Data and Tracks

3.1 Source Data and Label Algebra

The task starts from the reversible subset of PhrasIS. The English source data contains 1,946 human-annotated source pairs and 3,892 ordered instances after materializing both directions. The three reversible labels are close to balanced (Table 1), reducing the risk that paired correctness can be inflated by a trivial majority-class strategy.

The official label inventory contains four code values: EQUIVALENCE, FORWARD_ENTAILMENT, BACKWARD_ENTAILMENT, and NEGATIVE_OTHER. The first three labels form the reversible algebra. Original PhrasIS labels outside this algebra will be pooled as NEGATIVE_OTHER. These items make the classification problem less artificially clean and help systems distinguish directional entailment from weaker or unrelated semantic relations. They enter item-level label evaluation but are excluded from SOFTCONS and HARDCONS, which are defined only over reversible source pairs.

Track	Name	Description
1	English	Monolingual English phrase-pair NLI
2	Spanish	Monolingual Spanish phrase-pair NLI
3	Basque	Monolingual Basque phrase-pair NLI
4	Mixed	Premise and hypothesis use different EN/ES/EU languages

Table 2: Planned DiCo-NLI tracks.

NEGATIVE_OTHER items will be capped and stratified so that they do not dominate the reversible labels; final counts and per-split distributions will be documented for every track.

3.2 Tracks and Scale

DiCo-NLI will include four independent submission tracks (Table 2). Participants may enter any subset of tracks. The three monolingual tracks evaluate standard within-language phrase-pair NLI. Cross-lingual transfer is handled explicitly in the Mixed track, where premise and hypothesis may use different languages. This design avoids ambiguity about whether EN–EU transfer is hidden inside the monolingual tracks and makes macro-averages interpretable.

Before quality filtering, the reversible component yields up to 11,676 monolingual ordered instances across EN/ES/EU and 23,352 mixed-language ordered instances, for 35,028 reversible ordered instances across all tracks. Final counts will be lower if translation validation removes unstable target-language pairs. If filtering distorts a target-language label distribution, controlled stratified downsampling will restore comparable proportions and the adjustment will be reported.

3.3 Splits, Leakage, and Contamination

The official split unit is the underlying source phrase pair, not the ordered instance. For every source pair (A, B) , all derived records are assigned to the same train, development, trial, or evaluation partition: the original direction, the reversed direction, Spanish and Basque translations, and every mixed-language variant. Thus, if a source pair appears in the final evaluation split, no direction, language, or variant of that pair appears in released training or development data. This implements the hard/unseen setting from the LREC pilot as the official SemEval design.

We do not claim that DiCo-NLI is contamination-free. PhrasIS has been publicly described since 2024, and frontier models may have seen some examples or documentation. The task nevertheless reduces avoidable leakage by using new grouped

splits, hidden final labels during evaluation, delayed gold release, complete provenance metadata, newly produced multilingual variants, and exact/near-exact overlap audits against examples appearing in papers, public documentation, and released sample material. The official data release will include a `data/README.md` documenting split construction, source-pair grouping, label distributions, provenance, and contamination-control decisions.

3.4 Multilingual Production and Validation

The multilingual tracks require more than phrase-level translation fidelity: the translated pair must preserve the intended NLI relation and its deterministic reverse relation. Spanish and Basque candidates will be generated independently by multiple systems. The initial generators are NLLB-200 and Google Cloud Translation, representing an open multilingual model and a production API. For Basque, we will also use the most recent Latxa version available through a HiTZ API. During internal preparation, we will additionally try selected open and closed translation-capable LLMs through OpenRouter or comparable providers when licensing, cost, and reproducibility constraints permit. System/API versions, providers, prompts, decoding settings, and dates will be frozen and recorded. Generators receive only the phrases, without relation labels or reverse metadata, so that translation is not biased toward the expected label.

Automatic agreement signals will be used only for triage. High-disagreement cases will be inspected more carefully, but automatic overlap is not treated as evidence that an entailment relation is preserved. For each target language, two bilingual annotators will independently assess a 100-source-pair pilot audit. The audit will check: (i) phrase-level translation adequacy, (ii) preservation of the original NLI label, and (iii) preservation of the reverse label after swapping premise and hypothesis. The audit will deliberately include, where available, examples involving quantification, plurality, definiteness, specificity, genericity, scope, case marking, clitics, and other morphosyntactically sensitive phenomena. We will report raw agreement, Cohen’s κ , adjudication counts, and translation-induced label-flip or instability rates.

For full production, one bilingual reviewer checks every translated item. A second reviewer adjudicates low-confidence, ambiguous, or relation-changing cases. Items are corrected when a

relation-preserving translation can be found; otherwise they are removed from official evaluation. This applies equally to apparently simple examples such as *Everyone is hungry* / *Someone is hungry*, because quantifier scope and generic/specific readings can be destabilized by translation.

3.5 Release, Licenses, and Ethics

The upstream PhrasIS repository is MIT-licensed; its notice and provenance will be retained. Newly authored translations, adjudications, splits, and metadata will be released under CC BY 4.0, while scorer and starter-kit software will use GPL-3.0. A component-level license manifest, GitHub repository, and permanent Zenodo record will accompany trial and final data. Before release, final items will be screened for private-person personally identifiable information and sensitive content. The task supports semantic classification research and involves neither personal profiling nor medical decision making. The main methodological risk is benchmark overfitting to shallow cues, which is one of the behaviors the task is designed to expose.

4 Pilot Evidence, Trial Data, and Baselines

4.1 Pilot Results

The task is grounded in the public LREC 2026 pilot (Apalaza Larraya et al., 2026), which evaluated encoder-only, decoder-only, and encoder-decoder systems on English PhrasIS reversal pairs. The pilot included DeBERTa-v3, ModernBERT, RoBERTa, OPT, Pythia, GPT-2, BART, and Flan-T5. Table 3 gives representative and family-level results. Weighted F_1 is measured on the positive combined English test track; SOFTCONS and HARDCONS use the unseen/hard positive combined track, the condition adopted for SemEval.

These results demonstrate feasibility and leave room for improvement. Encoder-only systems perform strongest overall, but even the best representative encoder does not solve the hard/unseen directional setting. Decoder-only systems lag substantially, especially under paired correctness, and encoder-decoder systems show a different profile: their SOFTCONS can be close to encoder-only models even when item-level F_1 is lower. This supports reporting item-level and paired metrics together rather than collapsing evaluation to a single coarse score.

Family	Representative model	Representative $F/S/H$	Family mean $\pm$ sd $F/S/H$
Encoder-only	DeBERTa-v3-base	.75/.84/.79	.713 $\pm$.031 / .770 $\pm$.049 / .721 $\pm$.048
Decoder-only	OPT-125M	.61/.65/.58	.583 $\pm$.038 / .568 $\pm$.060 / .498 $\pm$.070
Encoder-decoder	BART-large	.68/.80/.72	.655 $\pm$.024 / .768 $\pm$.025 / .693 $\pm$.019

Table 3: Pilot results. F : weighted F_1 ; S : SOFTCONS; H : HARDCONS.

4.2 Trial Data and Supplementary Materials

For the revision package, we will provide pilot/trial data demonstrating the complete task format. The trial release will be a source-pair grouped random sample from PhrasIS, including original and reversed records, labels, reverse identifiers, split metadata, source provenance, and language fields. The sample will exercise all four labels in the official format. If Spanish or Basque validation is not complete by the revision deadline, the trial data will clearly mark which records are English-only pilot records and which multilingual records are provisional examples.

The supplementary upload will also include baseline result sheets confirming that the evaluation code is operational. The LREC pilot paper will be attached as the detailed baseline evidence for the broad model comparison, and the supplementary results will expose the compact table above in machine-readable form. The evaluation plan will be provided as a separate document and will match the metric definitions in Section 5.

4.3 Starter Kit

The starter kit is intended to provide runnable reference systems, not the strongest possible systems. We will fix one reference per architecture family: microsoft/deberta-v3-base for encoder-only classification, facebook/opt-125m for decoder-only classification or constrained generation, and facebook/bart-base for encoder-decoder generation/classification. For each baseline, we will release preprocessing code, configuration files, prediction format, scorer command, learning rate, batch size, epochs, maximum sequence length, optimizer, random seeds, model-selection rule, and trained checkpoints when redistribution permits. Data construction and official scoring are CPU-based; each starter baseline will use a documented one-GPU configuration.

5 Evaluation

5.1 Official Scores

Systems predict one label for every ordered phrase pair. The official evaluation for each track reports three main scores: weighted F_1 over all four labels, SOFTCONS, and HARDCONS. These scores are not collapsed into a single primary metric because they answer different questions. Weighted F_1 measures item-level label correctness, SOFTCONS measures directional self-consistency independently of correctness, and HARDCONS measures paired directional correctness over reversible source pairs. Macro- F_1 and per-label F_1 are also reported as secondary diagnostics. Official results are track-specific. We will additionally report a monolingual macro-average over EN/ES/EU and an all-track macro-average over EN/ES/EU/Mixed, but neither replaces the per-track leaderboards.

The scientific analysis will foreground the full weighted- F_1 /SOFTCONS/HARDCONS profile: a system may be accurate but directionally unstable, internally stable but systematically wrong, or strong only when both directions of a reversible pair are considered together. HARDCONS is interpreted as a strict paired-correctness summary rather than as an independent consistency axis beyond correctness. If the CodaBench interface requires a single display ordering or an additional summary field, that convention will be preannounced in the scorer documentation and will not replace the three official evaluation scores.

5.2 Metric Definitions

Let L be the full label set, n_ℓ the support of label ℓ , and $N_L = \sum_{\ell \in L} n_\ell$. For each label ℓ ,

$$P_\ell = \frac{TP_\ell}{TP_\ell + FP_\ell}, \quad R_\ell = \frac{TP_\ell}{TP_\ell + FN_\ell},$$

$$F_\ell = \frac{2P_\ell R_\ell}{P_\ell + R_\ell}, \quad F_w = \sum_{\ell \in L} \frac{n_\ell}{N_L} F_\ell.$$

The scorer will follow standard zero-division handling and will document this behavior in the released evaluation plan.

Let $\mathcal{P}$ be the set of reversible source pairs in a track. For pair $i \in \mathcal{P}$, let x_i be the original ordered item, x_i^r its reversed counterpart, y_i the gold label for x_i , $\hat{y}_i = f(x_i)$ the system prediction, and $\hat{y}_i^r = f(x_i^r)$ the prediction for the reversed item. The deterministic reversal operator Rev maps EQUIVALENCE to itself, FORWARD_ENTAILMENT to BACKWARD_ENTAILMENT, and BACKWARD_ENTAILMENT to FORWARD_ENTAILMENT. NEGATIVE_OTHER is not included in $\mathcal{P}$.

$$s_i = \mathbf{1}[\hat{y}_i^r = \text{Rev}(\hat{y}_i)],$$

$$\text{SOFTCONS} = |\mathcal{P}|^{-1} \sum_{i \in \mathcal{P}} s_i.$$

SOFTCONS measures directional compatibility independently of correctness. For example, predicting EQUIVALENCE in both directions is soft-consistent even if the correct labels are FORWARD_ENTAILMENT and BACKWARD_ENTAILMENT. Predicting FORWARD_ENTAILMENT in both directions is not soft-consistent.

$$h_i = \mathbf{1}[\hat{y}_i = y_i] \mathbf{1}[\hat{y}_i^r = \text{Rev}(y_i)],$$

$$\text{HARDCONS} = |\mathcal{P}|^{-1} \sum_{i \in \mathcal{P}} h_i.$$

HARDCONS is paired directional correctness. Because the reverse gold label is deterministic, correctness in both directions entails consistency automatically. Thus, HARDCONS is paired accuracy over reversal-defined semantic units. It is still useful because it changes the unit of evaluation from isolated ordered instances to complete reversible source pairs: two systems can obtain the same item-level score while differing in whether they solve both directions of the same relation.

5.3 Run Metadata and Rules

Run metadata is mandatory. Participants will report access regime (open-weight, commercial/API, private/closed, or hybrid/ensemble), architecture family, model/API identifier and version, model size when known, external data use, retrieval, prompting or fine-tuning strategy, and multilingual strategy. Access regime supports post-task analysis and does not define a separate leaderboard. Verification will be partial and based on system papers, released code/configurations, model/API documentation, and organizer sanity checks; unverifiable

fields will be marked undisclosed. Participants may use external resources unless the final rules restrict a specific resource, but all such use must be documented.

6 Infrastructure, Schedule, and Organizers

The task will run on CodaBench with a public repository containing trial data, scorer, format checker, starter kit, submission examples, and documentation. Final gold labels will be hidden during the official evaluation phase and released afterward together with the scorer and data documentation. Task artifacts will be archived on Zenodo for long-term access.

We will follow the SemEval-2027 schedule. Sample data is due by 15 July 2026, training data by 1 September 2026, internal evaluation data by 1 December 2026, official evaluation in January 2027, system papers in February 2027, notifications in March 2027, camera-ready papers in April 2027, and the SemEval workshop in summer 2027. The three-week revision period will be used to prepare the revised proposal, trial/pilot sample, baseline result sheets, evaluation plan, and supporting documentation in a private development repository before public release.

Organizers. **Inigo Lopez-Gazpio** (lead) works on semantic evaluation, STS, fine-grained semantics, and benchmark design, and co-organized SemEval-2015 Task 2, SemEval-2016 Task 2, and SemEval-2017 Task 1. **Jon Felix Apaolaza Larraya** (co-organizer) led the DiCo-NLI pilot and will coordinate data, scoring, and analysis. **Aitor Soroa** (advisory) contributes expertise in semantic processing and robust evaluation. **Rodrigo Agerri** (advisory) contributes multilingual NLP and system evaluation expertise. All organizers are affiliated with the HiTZ Basque Center for Language Technology–Ixa NLP Group, University of the Basque Country UPV/EHU. The team will maintain participant communication and infrastructure, release data and software, manage submissions and reviewing, and write the task description paper.

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SemEval Task Proposals 2027 Reviews for Submission #20

Title: SemEval-2027 Task Proposal: Coherence-Aware Compositional Fine Grained Natural Language Inference (CoCo-NLI) Authors: Jon Felix Apaolaza Larraya, Aitor Soroa, Rodrigo Agerri and Inigo Lopez-Gazpio

META-REVIEW

Comments:

Overall, reviewers are positive about this proposal. The task appears to be novel and interesting; data and pilot experiments exist; reviewers believe the proposal is well-structured and evaluation setup is strong; organizing team are strong too. Both reviewers expressed some concerns over the conceptual framing of the task and some other technical details, but these do not undermine the strengths of the submission.

REVIEWER #1

Detailed Comments

This proposal takes the reversal curse and reframes it as a paired fine-grained NLI task on PhrasIS, with a deterministic label-reversal operator and two coherence-based metrics (SoftCoh, HardCoh) on top of weighted F1, across EN/ES/EU.

The team is strong, the pilot exists, the infrastructure plan is credible, and the evaluation section is the cleanest part of the proposal. I want to like this more than I do, and I'll try to explain why.

(R1.1) My main concern is conceptual. The reversal curse as introduced by Berglund et al. is a claim about parametric knowledge: the model has seen “A is B” in training and fails to produce “B is A” at inference. It’s a failure of weight-level symmetry. What CoCo-NLI measures is something different: given both phrases in context, does the model assign mutually compatible directional labels when the pair is flipped? That’s in-context compositional consistency, and it’s a perfectly worthwhile thing to measure, but it is not the same phenomenon.

Answer R1.1 We agree with the reviewer that DiCo-NLI is not identical to the original Reversal Curse setup of Berglund et al., which targets parametric factual reversal: a model exposed to a statement such as A is B fails to retrieve or generate the reverse form B is A. DiCo-NLI instead operates at the level of fine-grained NLI relation prediction.

However, the task is closer to the Reversal Curse motivation than a purely test-time pair-flipping consistency check. In our LREC pilot, we explicitly distinguished two directional evaluation regimes. In the **seen/easy** condition, models were fine-tuned/model-selected on the original directional constructs and then evaluated on their reversed manipulated counterparts. This tested whether exposure to one directional formulation supported correct behavior under reversal. In the **unseen/hard** condition, both the original and reversed phrase-pair constructs came from the held-out evaluation split, testing directional generalization to novel pairs.

The LREC experiments showed that the unseen/hard setting was substantially more challenging than the seen/easy setting across model families. For the SemEval task, we therefore decided to focus the official evaluation directly on this harder and cleaner scenario. This keeps the task simpler for participants, avoids introducing an additional seen/unseen split distinction in the competition setup, and makes the benchmark target clearer: systems must generalize directional NLI decisions to held-out phrase pairs and their reversed counterparts.

We have revised the framing to avoid claiming that the task directly measures Berglund-style parametric knowledge reversal. The task is now presented as **DiCo-NLI: Directional Consistency in Fine-Grained Natural Language Inference**. Berglund et al. motivate the use of reversal as a stress test for directional generalization, while DiCo-NLI operationalizes that idea in a supervised fine-grained NLI setting through a deterministic label-reversal operator.

(R1.2) A model can be perfectly consistent on CoCo-NLI and still fail Berglund-style probes, or vice versa. The proposal elides this (“the inability of LLMs... to preserve inference when relation direction is reversed”) and uses the gravity of the RC literature to motivate a task that doesn’t actually test the RC.

Answer R1.2 We agree that DiCo-NLI scores and Berglund-style Reversal Curse scores are not logically interchangeable at the level of individual systems. A model may be consistent when both phrases are provided in an NLI input pair and still fail a parametric factual-reversal probe; conversely, a model may retrieve reversed factual associations while still failing to assign fine-grained entailment labels consistently. We therefore do not treat DiCo-NLI as a direct proxy measurement of the original Reversal Curse.

However, the connection to Berglund et al. is not merely rhetorical. The closest point of contact is the **seen/easy** condition in our LREC pilot. In that setting, models were fine-tuned/model-selected on original directional phrase-pair constructs and then evaluated on their reversed manipulated counterparts. This is structurally analogous to Berglund et al. in the sense that exposure to one directional formulation is tested against behavior in the reverse direction. The crucial difference is the target of prediction: Berglund et al. evaluate parametric factual recall/generation, whereas DiCo-NLI evaluates fine-grained NLI relation prediction.

Berglund et al. Reversal Curse	LREC seen/easy DiCo-NLI setup
Model is trained/fine-tuned on A is B	Model is fine-tuned/model-selected on an original directional phrase construct A -> B
Model is queried on the reverse form B -> A	Model is evaluated on the reversed manipulated phrase pair B -> A
Tests whether one directional exposure supports reverse factual retrieval	Tests whether one directional NLI construct supports reverse-label prediction
Output is factual recall or generation	Output is a fine-grained NLI label

The SemEval task itself focuses on the harder **unseen** condition, because our LREC experiments showed that this setting is more challenging and cleaner for participants: both the original and reversed phrase-pair constructs are held out from training. This design choice avoids adding a separate seen/unseen competition distinction while retaining the core reversal-based diagnostic.

We have revised the motivation accordingly. Berglund et al. are now presented as conceptual and methodological motivation for reversal-based evaluation, not as a claim that DiCo-NLI directly measures weight-level factual reversal. DiCo-NLI contributes to the broader line of work on directional generalization by operationalizing reversal at the level of fine-grained NLI labels and deterministic label reversal.

(R1.3) I’d push the authors to either (a) rename and remotivate the task as directional inferential coherence, which is what it actually is! and which I think is defensible on its own terms, or (b) include a component that correlates CoCo-NLI performance with a parametric-reversal probe and argues empirically that the two are connected.

Answer R1.3 We follow the reviewer’s recommendation and adopt option (a): the task has been renamed and remotivated around directional consistency rather than presented as a direct Reversal Curse benchmark. The revised task name is **DiCo-NLI: Directional Consistency in Fine-Grained Natural Language Inference**.

We intentionally use the term **consistency** rather than **coherence**. Although the reviewer’s phrase “directional inferential coherence” captures the intended evaluation target, “coherence” has a well-established meaning in discourse analysis and may suggest discourse-level textual organization, discourse connectedness, or interpretive continuity across larger text units. During discussions of our LREC work, linguists also pointed out that using “coherence” for our label-level reversal metric could be misleading from a discourse-analysis perspective. This feedback made clear that “consistency” is the more precise term for the property we evaluate: whether predictions over a phrase pair and its reversed counterpart remain mutually compatible under a deterministic label-reversal operator.

We do not plan to add a separate parametric-reversal probe as an official component of the SemEval task. Such a component would require a different experimental design, including controlled factual exposure, factual-recall prompts, and assumptions about whether systems have or have not seen particular facts during training. This would blur the scope of the shared task and make the evaluation harder to interpret for participants. Instead, the official task focuses on a single, well-defined problem: fine-grained NLI label prediction under directional reversal.

We agree that the empirical relationship between DiCo-NLI and Berglund-style parametric reversal is an interesting research question. We therefore now present it as a possible follow-up analysis rather than as an assumption built into the shared-task definition.

(R1.4) Second concern, on the metric. **HardCoh** is defined as coherent-under-Rev AND both predictions correct. But Rev is deterministic and the gold is consistent under Rev by construction. So if both predictions are correct, coherence is automatic. **HardCoh** therefore reduces to paired accuracy “both items in the minimal pair labeled correctly” which is a stricter accuracy variant, not a distinct evaluation axis.

Answer R1.4 We agree with the reviewer’s mathematical observation. Because the reversed gold label is generated by a deterministic reversal operator, correctness in both directions entails consistency automatically. Thus, for the reversible subset, **HardCons** should be understood as **paired directional correctness**: a pair receives credit only when both the original item and its reversed counterpart are predicted correctly.

We have revised the terminology and explanation accordingly. We no longer present the hard metric as an independent consistency condition beyond correctness. Instead, we describe it as the strict paired counterpart to item-level label evaluation. This distinction is still important because **HardCons** is not equivalent to ordinary item-level accuracy or F-measure.

For example, consider two reversible pairs, i.e. four ordered instances:

Pair	System A: original correct?	System A: reverse correct?	System B: original correct?	System B: reverse correct?
1	yes	no	yes	yes
2	no	yes	no	no

Both systems obtain the same item-level accuracy:

System A accuracy = $2/4 = 0.50$

System B accuracy = $2/4 = 0.50$

However, their paired hard scores differ:

System A **HardCons** = $0/2 = 0.00$

System B **HardCons** = $1/2 = 0.50$

System A gets one direction correct in each pair, but never solves a complete reversible pair. System B solves one reversible relation in both directions. Thus, **HardCons** captures whether correctness is paired under reversal, rather than merely accumulated over isolated ordered instances.

More generally, if item-level correctness were independent with probability p , the expected paired hard score would be p^2 , not p . This illustrates why the metric is stricter than item-level accuracy: partial success on one direction of a reversible relation is not enough.

In the revised task description, we therefore report **F-measure** for standard label prediction, **SoftCons** for self-consistency independent of correctness, and **HardCons** for paired directional correctness over reversible instances.

(R1.5) **SoftCoh** is the metric doing real work: it separates self-consistency from correctness and can catch a model that’s reliably wrong in a coherent way. I think the paper would be stronger if it led with the F1 vs **SoftCoh** gap as the diagnostic signature and presented **HardCoh** honestly as paired accuracy. As written, **HardCoh** is oversold.

Answer R1.5 We agree with the reviewer that **SoftCons** is the metric that most directly isolates the novel diagnostic dimension of the task. **F-measure** evaluates whether individual ordered instances are labeled correctly, whereas **SoftCons** evaluates whether a model’s predictions are mutually compatible under reversal, independently of whether those predictions are correct. The contrast between these two metrics is therefore central: a model may obtain reasonable label-level performance while still producing directionally inconsistent predictions, or it may be internally consistent but systematically wrong.

We have revised the task description to foreground this point. In the updated framing, **F-measure** is the standard label-prediction metric, **SoftCons** is the main diagnostic metric for directional self-consistency, and **HardCons** is described more conservatively as paired directional correctness over reversible instances.

We also agree that the previous wording oversold the hard metric. Since the reversed gold label is deterministic, **HardCons** is not an independent consistency axis beyond correctness; it is a strict paired-correctness measure. We keep it because it answers a different practical question from **F-measure**: whether a system solves both directions of a reversible relation as a unit. However, the revised proposal will no longer present **HardCons** as conceptually independent from correctness. Instead, the task analysis will emphasize the joint interpretation of the three scores:

- **F-measure**: item-level label correctness;

- **SoftCons**: prediction-level directional consistency, regardless of correctness;
- **HardCons**: paired correctness under reversal.

This revision should make the diagnostic role of **SoftCons** explicit while preserving **HardCons** as a useful stricter evaluation summary.

(R1.6) Third, and this is the one that most affects my score: there are no numbers. The proposal lists eight baseline architectures, cites a pilot, asserts there is clear room for improvement, and then asks reviewers to take all of that on faith while the pilot paper is in press and not accessible. A compact three-row table one encoder, one decoder, one enc-dec, with F1/SoftCoh/HardCoh would have made the case concrete and would have let me verify the “room for improvement” claim myself. Its absence is particularly awkward given that the whole argument for running the task is that current systems fail on it.

Answer R1.6 We agree with the reviewer. The proposal should have included a compact baseline table rather than relying on a citation to an in-press pilot paper. At the time of submission, the LREC paper was not yet externally available, but this does not remove the need to provide enough empirical evidence inside the proposal itself. The pilot paper is now available here: <https://lrec.elra.info/lrec2026-main-423>

We have now added representative pilot results to the task documentation and will include them in the revised proposal/task description. We include two summaries below: first, one representative high-performing model from each architecture family; second, the mean and sample standard deviation across all evaluated models in each family. This avoids relying on a single cherry-picked run while keeping the table compact. **F-measure** corresponds to standard label prediction on the positive combined test track. **SoftCons** and **HardCons** correspond to the unseen/hard directional-consistency evaluation on the positive combined track, which is the setting closest to the planned SemEval evaluation.

Representative models:

Architecture family	Representative model	F-measure	SoftCons	HardCons
Encoder-only	DeBERTa v3-base	0.75	0.84	0.79
Decoder-only	OPT-125M	0.61	0.65	0.58
Encoder-decoder	BART-large	0.68	0.80	0.72

Family-level mean $\pm$ sample standard deviation:

Architecture family	F-measure	SoftCons	HardCons
Encoder-only	0.713 ± 0.031	0.770 ± 0.049	0.721 ± 0.048
Decoder-only	0.583 ± 0.038	0.568 ± 0.060	0.498 ± 0.070
Encoder-decoder	0.655 ± 0.024	0.768 ± 0.025	0.693 ± 0.019

These results support the feasibility of the task while also showing clear room for improvement. Encoder models perform strongest overall, but even the best representative encoder does not solve the hard/unseen directional setting. Decoder-only models lag substantially, especially under **HardCons**, and encoder-decoder models are competitive but still below the strongest encoders. The family-level means show that these are not isolated model choices: architecture family matters systematically in the pilot.

We will release the corresponding baseline scripts, documented hyperparameters, and scorer as part of the starter kit. At minimum, the starter kit will include one tuned encoder baseline and one decoder or encoder-decoder baseline; our goal is to include representative models from all three architecture families.

A few more things worth raising.

(R1.7) Contamination is not discussed. PhrasIS has been public since 2024 and any frontier model trained through 2025-26 has plausibly ingested it.

Answer R1.7 We agree that contamination must be treated explicitly. PhrasIS has been described in the literature since 2024, and we cannot assume that large frontier models have not been exposed to some form of the resource or to examples from the paper. We will therefore not present the benchmark as contamination-free by default.

At the same time, the official DiCo-NLI evaluation set is not simply the public PhrasIS dataset reused as-is. The SemEval data will be a task-specific construction: we select the reversible labels, generate paired reversed instances through the deterministic reversal operator, add **NEGATIVE_OTHER** samples from the non-reversible PhrasIS labels as noise, create new Spanish and Basque versions for the task, and define new trial/final splits. The multilingual and mixed-language tracks in particular will be newly produced for the shared task and will not correspond to a pre-existing public benchmark.

To reduce contamination risk, we will document the split construction in the **data/** folder and apply several safeguards: hidden final test labels during the competition, delayed release of gold labels, a held-out final evaluation split generated after task specification, provenance metadata for all items, and an audit of exact or near-exact overlap with examples that have appeared in papers or public documentation.

Finally, we will make the contamination risk explicit in the task documentation. Since DiCo-NLI evaluates not only item-level **F-measure** but also paired direc-

tional consistency over reversed instances, memorization of isolated labels is less sufficient than in a standard static classification benchmark. Nevertheless, we agree with the reviewer that contamination can affect leaderboard interpretation, and we will report the construction protocol and contamination-control decisions together with the official release.

(R1.8) For a task whose scientific payload is measuring an LLM reasoning failure, the proposal needs at least a paragraph on how the test split is constructed unreleased items, newly written, perturbed, canary-checked, something.

Answer R1.8 This concern is related to contamination but concerns the concrete construction of the final evaluation split, so we address it explicitly. The official DiCo-NLI test set will be built as a held-out grouped split, not as a random list of ordered instances.

The grouping unit will be the underlying source phrase pair. For every source pair (A,B), all derived instances will be assigned to the same split: the original ordered item (A,B), the reversed item (B,A), the corresponding Spanish and Basque translations, and any mixed-language variants derived from that source pair. This prevents leakage where a model could see one direction, language, or variant of a pair during training and be evaluated on another variant during testing.

Concretely, if a reversible pair appears in the final test set, neither its original direction nor its reversed counterpart will appear in the released training or development data. Likewise, multilingual versions of that same source pair will not be split across train/dev/test. This implements the hard/unseen setting from our LREC pilot as the official SemEval setup.

We will also include **NEGATIVE_OTHER** items from non-reversible PhrasIS labels as noise in the label-prediction evaluation. These items will be split using the same grouped protocol and will be excluded from **SoftCons** and **HardCons**, which are defined only over reversible pairs.

The final data release will include a **data/README.md** documenting the split construction, source provenance, label distributions, and the relation between source pairs and derived instances. Before release, we will audit exact and near-exact overlap with examples appearing in papers, task documentation, and public-facing material. Gold test labels will remain hidden during the competition and released only after the official evaluation period.

This protocol is designed to make the leaderboard reflect generalization to held-out directional NLI constructs rather than memorization of isolated phrase-pair labels.

(R1.9) The translation protocol is underspecified in a way that matters for this specific task.

“Semiautomatic translation by deep MT and bilingual annotators” doesn’t say

which system, how many annotators per item, or crucially, how the reversal operator is validated after translation. Basque case marking and Spanish clitic/scope interactions can plausibly shift entailment direction on short phrase pairs.

Answer R1.9 We agree that the original proposal underspecified the translation protocol. For this task, translation quality cannot be evaluated only as phrase-level fidelity: the translated pair must also preserve the intended directional NLI relation and its reversed counterpart under the label-reversal operator.

We have therefore revised the multilingual data protocol as follows. First, candidate Spanish and Basque translations will be generated using multiple independent translation systems, including neural MT and LLM-based translation. The exact systems used will be documented in the data release. Each phrase will be translated without exposing translation systems to the source-side relation metadata, so that the translation step itself is not biased toward the expected label.

Second, automatic agreement between candidate translations will be used as a triage signal, not as the final validation criterion. We will compute surface and normalized overlap signals, such as token/lemma Jaccard or related string-similarity measures, to identify high-disagreement cases that require closer inspection. However, we agree that such automatic scores are insufficient for Spanish and Basque, where morphology, case marking, clitics, and scope can affect the resulting relation.

Third, validation will be performed by bilingual annotators at the phrase-pair level. We will have two bilingual annotators per target language. For each language, both annotators will independently validate a pilot audit of 100 source pairs. This audit will check: (i) phrase-level translation adequacy, (ii) preservation of the expected NLI label, and (iii) preservation of the deterministic reversed label after swapping premise and hypothesis. We will report raw agreement and Cohen’s kappa on this pilot validation, together with the proportion of translation-induced label flips or unstable cases.

For the full release, each translated item will be reviewed by one bilingual annotator. Items flagged as low-confidence by automatic agreement scores, judged ambiguous by the first annotator, or suspected of changing the directional relation will be reviewed or adjudicated by the second annotator. A translated pair will only be retained if the original and reversed target-language instances remain compatible with the expected label-reversal mapping.

If the problem is a bad translation, the item will be corrected. If the target-language construction naturally changes or destabilizes the entailment relation, the item will be removed from the official evaluation split rather than silently kept with the English-derived label.

Finally, the data documentation will report the translation systems used, the validation protocol, annotator agreement on the pilot audit, adjudication counts, and translation-induced label-flip statistics. This makes the multilingual ex-

tension auditable and ensures that the Spanish and Basque tracks evaluate directional consistency rather than artifacts of translation.

(R1.10) The “Everyone is hungry / Someone is hungry” example is fine in English under a generic reading, but generic/specific scope is exactly the kind of thing that translation destabilizes. With 200 expert hours budgeted for ES+EU combined over roughly 4k 2 items, I’d want the protocol pinned down and a translation-induced label-flip audit on a pilot sample.

Answer R1.10 We agree with the reviewer that the “Everyone is hungry / Someone is hungry” example is deceptively simple. Quantification, generic/specific readings, plurality, definiteness, and scope are precisely the types of phenomena that can be destabilized by translation, especially across typologically different languages such as English, Spanish, and Basque.

We therefore will not assume that short phrase translations are automatically relation-preserving. Instead, the validation protocol described in R1.9 explicitly treats translation as a label-preservation problem. The translated target-language pair must preserve both the original NLI label and the deterministic reversed label. Quantifier- and scope-sensitive pairs will be treated as high-priority cases in the translation audit.

Concretely, the 100-source-pair pilot audit for each target language will include, where available, examples involving quantification, plurality, specificity/generality, determiners, and related scope-sensitive constructions. The audit will estimate how often such items preserve the English label after translation and how often they induce a label flip or unstable interpretation. We will report these counts together with raw agreement and Cohen’s kappa.

If a translated item changes the entailment direction or becomes ambiguous in the target language, it will not be kept with the English-derived label. The item will either be corrected by choosing a better target-language formulation or removed from the official evaluation split. This applies equally to apparently simple examples such as **Everyone is hungry / Someone is hungry**.

Thus, the multilingual tracks will not rely on the assumption that translation is trivial. They will rely on explicit bilingual validation of directional label preservation.

(R1.11) The cross-lingual evaluation story is also ambiguous. Are the three tracks independent, or is EN-EU transfer a first-class submission? This affects how participants build systems and whether the macro-average is meaningful.

Answer R1.11 We agree that the original proposal did not define the multilingual evaluation structure clearly enough. We have revised the task description to make the tracks explicit.

The shared task will include four tracks:

Track	Description
English	Monolingual English premise-hypothesis pairs
Spanish	Monolingual Spanish premise-hypothesis pairs
Basque	Monolingual Basque premise-hypothesis pairs
Mixed multilingual	Premise and hypothesis may appear in different languages among EN/ES/EU

The three monolingual tracks are independent evaluation tracks. Participants may submit to any subset of them, and results will be reported separately per language. Cross-lingual transfer is not implicitly mixed into those tracks.

The cross-lingual setting will be evaluated explicitly through the fourth mixed multilingual track. This track is intended for systems that can handle language combinations across the premise and hypothesis, for example EN→ES, ES→EU, or EU→EN. This makes cross-lingual transfer a first-class evaluation condition without making it mandatory for teams participating only in monolingual tracks.

For reporting, we will prioritize per-track leaderboards. We may additionally report macro-averages, but only over clearly specified groups: a monolingual macro-average over EN/ES/EU and, separately, an overall macro-average including the mixed multilingual track if appropriate. This avoids ambiguity about what a macro-average represents and lets participants interpret their results according to the setting they targeted.

(R1.12) Minor: the access-regime tag (open-weight vs API) is a nice analytical addition but you don't say how it's verified or whether it's required.

Answer R1.12 We agree that the access-regime tag must be specified more clearly. In the revised task documentation, this information will be mandatory run metadata, but it will not define separate official leaderboards. The purpose is post-task analysis: we want to compare whether performance patterns differ across open-weight models, API-based systems, private closed models, and hybrid systems.

Participants will be required to report the access regime of each official run in their submission metadata and system description paper. The planned categories are: **open-weight**, **commercial/API-based**, **private/closed**, and **hybrid/ensemble**. We will also ask teams to report architecture family, model size when known, prompting/fine-tuning strategy, multilingual capability, and external data use.

Verification will necessarily be partial, especially for commercial/API-based systems. We will therefore not claim full independent verification of all access-

regime tags. Instead, the tag will be checked against the system description paper, model names, API provider information, released code/configuration files where available, and organizer sanity checks. Inconsistent or missing metadata may lead to a run being marked as **undisclosed** or excluded from access-regime-based analysis.

The official ranking will remain based on task metrics, not on access regime. The access-regime tag is an analytical variable intended to make the overview paper more informative and to avoid conflating very different system types in the post-task discussion.

(R1.13) The reference to Apaolaza et al. (2026) as in-press limits what reviewers can check, and for a proposal that leans this heavily on the pilot, a preprint link or appendix would have helped.

Answer R1.13 We agree. The proposal relied too heavily on an in-press pilot paper without giving reviewers enough access to verify the empirical basis of the task. We should have included either a preprint link, an appendix, or a compact table of pilot results in the proposal itself.

This has now been corrected. The LREC paper is publicly available here: <https://lrec.elra.info/lrec2026-main-423>. We have also added representative baseline results directly to the response/task documentation, including one model from each architecture family and family-level mean and standard deviation values. This ensures that the feasibility and difficulty of the task can be assessed without relying only on an inaccessible citation.

In the revised proposal and repository documentation, the pilot study will be linked explicitly, and the task description will include the relevant metric definitions, baseline summaries, and the relation between the LREC hard/unseen evaluation setting and the planned SemEval setup.

To be clear about what I do like: the paired-instance construction with a deterministic reversal operator is a genuinely good design choice, the formal definitions in the evaluation section are clean, HiTZ has a real track record on this kind of benchmark, and the Basque track is the sort of multilingual contribution the field claims to want and rarely gets.

If the conceptual framing is sharpened and a baseline table appears in the camera-ready, this is a solid shared task.

Questions for Authors

Is CoCo-NLI measuring the reversal curse (parametric, weight-level) or directional inferential coherence (in-context, compositional)?

Answer DiCo-NLI measures directional consistency in fine-grained NLI. It is not a direct measurement of Berglund-style parametric, weight-level knowledge reversal. We have revised the framing and renamed the task accordingly: **DiCo-NLI: Directional Consistency in Fine-Grained Natural Language**

Inference. Berglund et al. are used as conceptual and methodological motivation for reversal-based evaluation, not as a claim that the shared task directly measures the same parametric factual-reversal phenomenon.

Would you expect performance on the two to correlate, and if so, can you show it?

Answer We do not assume that performance on DiCo-NLI and performance on parametric-reversal probes must correlate at the level of individual systems. A model can plausibly succeed in one setting and fail in the other. However, our LREC pilot includes a seen/easy setting that is structurally close to Berglund et al.: models are exposed to one directional phrase-pair construct and evaluated on the reversed manipulated counterpart. The SemEval task focuses on the harder unseen setting, where both original and reversed constructs are held out. The relationship between DiCo-NLI scores and parametric reversal probes is therefore an empirical follow-up question, not an assumption of the task design.

Given that Rev is deterministic and gold labels are consistent under Rev, doesn't HardCoh collapse to paired accuracy?

Answer Yes, for the reversible subset, because the reversed gold label is deterministic, correctness in both directions entails consistency automatically. In that strict mathematical sense, **HardCons** is equivalent to paired accuracy over reversible pairs. We now describe it explicitly as **paired directional correctness** rather than as an independent consistency condition beyond correctness.

Its value is that the pairing is not arbitrary: it is defined by the label-reversal operator and by the original/reversed structure of the benchmark. **HardCons** changes the unit of evaluation from isolated ordered instances to reversible semantic units. A system receives credit only if it solves both directions of the same relation, and **NEGATIVE_OTHER** items are excluded because they do not have a deterministic reversed counterpart. Thus, although **HardCons** coincides numerically with paired accuracy in this setup, it operationalizes the task-specific notion of correctness under reversal and makes the comparison with **SoftCons** explicit: **SoftCons** asks whether predictions are directionally compatible regardless of truth, while **HardCons** asks whether the reversible relation is correctly solved in both directions.

If so, what does it add over F1 or paired accuracy that SoftCoh doesn't already provide?

Answer **HardCons** is not intended to replace **SoftCons**. **SoftCons** is the main diagnostic metric for directional self-consistency independent of correctness. **HardCons** adds a stricter paired-correctness view that ordinary item-level **F-measure** does not provide. Two systems can have the same item-level score while differing in whether they solve complete reversible pairs. This distinction also appears empirically in our pilot system results: architecture families do not collapse to the same profile under the three metrics. For example, encoder-decoder models obtain lower average **F-measure** than encoder-only models on

the positive combined track, but their average **SoftCons** is nearly identical, while decoder-only models drop much more sharply under **HardCons**. Thus the metrics expose different system behaviors rather than adding only redundant summaries. We therefore report the three metrics with distinct roles: **F-measure** for item-level label prediction, **SoftCons** for prediction-level directional consistency, and **HardCons** for paired correctness under reversal.

Can the camera-ready include a small baseline table at minimum one model per architecture family, reporting F1, SoftCoh, HardCoh?

Answer Yes. We have added representative baseline results from the LREC pilot and will include them in the revised task documentation. The LREC paper is now available at <https://lrec.elra.info/lrec2026-main-423>. We report one representative model per architecture family and family-level mean $\pm$ sample standard deviation for encoder-only, decoder-only, and encoder-decoder systems, using **F-measure**, **SoftCons**, and **HardCons**.

How is contamination handled? Is the test set held-out from public PhrasIS, newly authored, perturbed, or canary-checked?

Answer The official test set will be built as a held-out grouped split, not as a random list of ordered instances. The grouping unit is the underlying source phrase pair: original direction, reversed direction, Spanish/Basque translations, and mixed-language variants derived from the same source pair are assigned to the same split. Thus, if a pair appears in the final test set, neither its original nor reversed form appears in released train/dev data. We also add **NEGATIVE_OTHER** noise items from non-reversible PhrasIS labels, hide final test labels during evaluation, delay gold release, document provenance in **data/README.md**, and audit exact/near-exact overlap with public examples and documentation.

For ES/EU translation: which MT system, how many annotators per item, what’s the adjudication protocol, and have you audited how often translation flips the gold label on a pilot sample?

Answer The multilingual protocol now treats translation as label-preserving transfer, not only phrase-level translation. Candidate Spanish and Basque translations will be produced with multiple MT/LLM systems and automatically compared for triage. Two bilingual annotators per target language will independently validate a 100-source-pair pilot audit, checking translation adequacy, preservation of the NLI label, and preservation of the reversed label. We will report raw agreement, Cohen’s kappa, and translation-induced label-flip counts. For the full release, each item is reviewed by one bilingual annotator, with second-annotator adjudication for low-confidence, ambiguous, or relation-changing cases. Items that cannot preserve the intended relation are corrected or removed.

Are cross-lingual submissions (e.g., train EN, evaluate EU) part of the evaluation, or is each track independent?

Answer The task now has four explicit tracks: English, Spanish, Basque, and Mixed multilingual. The first three are independent monolingual tracks. Cross-

lingual transfer is handled explicitly in the Mixed multilingual track, where premise and hypothesis may appear in different languages among EN/ES/EU. Participants may submit to any subset of tracks. We will report per-track results and only compute macro-averages over clearly specified groups, such as monolingual EN/ES/EU or all four tracks.

How will the open-weight vs API tag be verified, and is it mandatory?

Answer The access-regime tag will be mandatory run metadata, but it will not define separate official leaderboards. Participants must report whether each run is **open-weight**, **commercial/API-based**, **private/closed**, or **hybrid/ensemble**, together with architecture family, model size when known, prompting/fine-tuning strategy, multilingual capability, and external data use. Verification will be partial and based on system papers, model/API names, released code/configuration files where available, and organizer sanity checks. Missing or inconsistent metadata may be marked as **undisclosed** or excluded from access-regime-based analysis.

REVIEWER #2

Detailed Comments

This is a well-motivated and carefully prepared proposal. CoCo-NLI targets a recognised gap in the current evaluation landscape for Large Language Models (LLMs): the inability of models to preserve logical inference when the direction of a relation is reversed, the so-called Reversal Curse. The proposal reframes this within a Natural Language Inference (NLI) setting, rather than the narrow factual-retrieval setup in which it was originally demonstrated, and extends it to a fine-grained, multilingual shared task.

The Reversal Curse has attracted substantial attention since Berglund et al. (2023) demonstrated that autoregressive LLMs trained on “A is B” systematically fail to infer “B is A.” Subsequent work has explored its theoretical underpinnings (NeurIPS 2024), proposed mitigation strategies such as reverse training Golovneva et al. (2024), and examined its manifestation across question-answering tasks Lin et al., (2024). Much of this work, however, has remained close to factual recall, question answering, or mechanistic analyses of reversal failures, rather than operationalising the phenomenon within a fine-grained NLI shared-task setting. To my knowledge, no prior shared task has combined the Reversal Curse framing with fine-grained directional NLI labels and a formal label-reversal coherence metric.

(R2.1) The closest related work in the NLI consistency literature is BECEL (Jang et al., COLING 2022), a benchmark that evaluates language model consistency across four logical properties, including symmetric consistency for NLI. However, BECEL is not a shared task, and its symmetric NLI test is conceptually distinct from CoCo-NLI’s design: BECEL evaluates symmetry only for restricted contradiction and selected neutral cases in standard 3-class

NLI, whereas CoCo-NLI explicitly models directional entailment reversal using MacCartney’s natural logic label-reversal operator, under which FORWARD-ENTAILMENT deterministically maps to BACKWARD-ENTAILMENT and EQUIVALENCE maps to itself. This is a richer and more formally grounded formulation than the symmetry setting explored in BECEL. More recent work on NLI meta-inferential properties (Blanck et al., arXiv 2026) and transitive self-consistency (Wu and Last, EMNLP 2025) similarly remains in the research-paper paradigm and does not directly address directional reversal coherence in a shared-task framework. The authors could strengthen the related work section further by citing these studies explicitly to sharpen the novelty argument.

Answer R2.1 We thank the reviewer for these references. We agree that they are directly relevant and will include them in the revised related work section to sharpen the novelty claim.

In particular, we will discuss BECEL as the closest benchmark in the NLI consistency literature. BECEL evaluates several logical consistency properties, including symmetric consistency for NLI, but its NLI setting is based on standard 3-class labels and restricted symmetry cases. DiCo-NLI differs by explicitly modeling directional entailment reversal with fine-grained Natural Logic-style labels: FORWARD-ENTAILMENT maps deterministically to BACKWARD-ENTAILMENT, while EQUIVALENCE maps to itself. Thus, the task is not only checking invariance under symmetry but evaluating direction-sensitive label reversal.

We will also cite Blanck et al. on NLI meta-inferential properties and Wu and Last on transitive self-consistency. These works are relevant because they study consistency properties of NLI systems beyond isolated accuracy. However, they remain research-paper benchmarks rather than shared tasks and do not define a multilingual fine-grained directional-reversal evaluation with an official scorer and participant submissions.

The revised related work will therefore position DiCo-NLI as complementary to this literature: it brings consistency-based NLI evaluation into a shared-task setting, focuses on fine-grained directional entailment rather than only 3-way NLI labels, and evaluates both item-level correctness and paired directional consistency across monolingual and mixed multilingual tracks.

The evaluation design is one of the strongest aspects of the proposal. The three-metric framework is well conceived:

- Weighted F: provides a standard and comparable label-prediction metric that connects the task to the broader NLI literature.
- SoftCoh isolates directional self-consistency from correctness, allowing the community to distinguish models that are internally coherent (even if incorrect) from those that produce incompatible predictions under reversal.
- HardCoh is the most task-aligned primary metric, as it rewards predictions

that are both correct and mutually consistent under the label-reversal operator.

The formal definitions of SoftCoh and HardCoh are clear and sufficiently specified for reproducible evaluation. The decision to report both per-language scores and a multilingual macro-average is also sensible. Finally, the plan to run the task on CodaBench with an official scorer and format validator is fully consistent with current SemEval best practices.

Data:

The data plan is realistic, though the risk profile differs between the English and multilingual tracks. The English track appears relatively low-risk from a data and infrastructure perspective: the pilot data already exists, the metrics are defined, and baseline systems have already been evaluated. The Spanish and Basque tracks, however, remain partly prospective. They require semi-automatic translation followed by bilingual expert adjudication, estimated at under 200 expert hours. While the resulting dataset of approximately 10k ordered instances across three languages is appropriate in scale, preserving directional entailment relationships across languages, particularly in Basque, whose morphosyntactic structure differs substantially from English and Spanish which introduces a moderate linguistic and organizational risk that the organizers will need to manage carefully.

The CC BY 4.0 license and commitment to long-term archiving on Zenodo are strong and fully aligned with current open-science expectations. The ethics section is brief but adequate: the data contains no personally identifiable information, no medical decision-making content, and no sensitive records.

The organizers also correctly identify benchmark overfitting to shallow cues as the primary methodological risk, which is precisely the behaviour the task is designed to expose.

Organizers:

The team of four organizers is well-suited to the task. The roles are clearly defined and the team has the institutional infrastructure.

Baselines

The pilot study is strong. Having already evaluated encoder-only models (DeBERTa-v3, ModernBERT, RoBERTa), decoder-only models (OPT, Pythia, GPT-2), and encoder-decoder models (BART, Flan-T5) on the English reversal-aware benchmark, the organizers are able to ground the task design in preliminary empirical findings rather than purely hypothetical motivations.

The pilot suggests several useful design lessons: (1) coherence should be treated as an explicit evaluation target rather than relying solely on aggregate label metrics; (2) the reversible entailment/equivalence subset appears to provide the clearest diagnostic signal; (3) encoder-based models performed more strongly in the reported experiments, although they still show confusion between entailment and high semantic similarity; and (4) performance appears sensitive to hyperparameter configuration.

The commitment to release baseline systems and training scripts should help lower the barrier to participation and improve reproducibility.

Suggestions for Improvement

While the proposal is strong overall, the following points deserve attention in the final task design:

(R2.2) Translation quality and linguistic risk. The proposal states that the Spanish and Basque tracks will be created through semi-automatic translation followed by bilingual annotator adjudication. It would be important to ensure that the adjudication process verifies not only translation fidelity, but also whether the logical coherence of the reversed pairs is preserved in the target languages. This is particularly relevant for Basque, whose morphosyntactic structure differs substantially from English and Spanish and may affect directional entailment relationships. A brief inter-annotator agreement protocol for the translated tracks would strengthen confidence in the multilingual extension.

Answer R2.2 We agree. This concern is closely related to the issues addressed in **R1.9** and **R1.10**. The multilingual extension must validate preservation of the directional NLI relation, not only translation adequacy. We have therefore revised the protocol so that Spanish and Basque annotation is treated as a label-preservation task.

Candidate translations will be generated using multiple MT/LLM systems, and automatic agreement signals such as token/lemma overlap will be used only for triage. The final decision will be made by bilingual annotators at the phrase-pair level. For each target language, two bilingual annotators will independently validate a pilot audit of 100 source pairs. The audit will check: (i) phrase-level translation adequacy, (ii) preservation of the expected NLI label, and (iii) preservation of the deterministic reversed label after swapping premise and hypothesis. We will report raw agreement, Cohen’s kappa, adjudication counts, and translation-induced label-flip rates.

For the full release, each translated item will be reviewed by one bilingual annotator. Items flagged as low-confidence, ambiguous, or relation-changing will be reviewed or adjudicated by the second annotator. If a translated pair

changes entailment direction or becomes unstable in the target language, it will be corrected if possible; otherwise, it will be removed from the official evaluation split.

This protocol directly addresses the linguistic risk noted by the reviewer, especially for Basque morphology and case marking, and ensures that the Spanish and Basque tracks evaluate directional consistency rather than artifacts of translation.

(R2.3) Label balance across languages. The proposal reports that PhrasIS contains 1,946 English source pairs with reversible labels, yielding approximately 4k ordered instances for the English track. It would be helpful to report the distribution of EQUIVALENCE, FORWARD-ENTAILMENT, and BACKWARD-ENTAILMENT labels in the pilot data. If the distribution is highly skewed, the organizers may wish to consider stratified sampling or related balancing strategies to avoid inflating HardCoh scores through majority-class behaviour.

Answer R2.3 We agree that label balance should be made explicit, both in the proposal response and in the released task documentation. The reversible English source subset used in the pilot contains 1,946 source phrase pairs and is close to balanced across the three reversible labels:

Gold label	Source pairs	Percentage
EQUIVALENCE	686	35.3
FORWARD_ENTAILMENT	624	32.1
BACKWARD_ENTAILMENT	636	32.7
Total	1,946	100.0

After adding the reversed counterpart for each source pair, the ordered-instance distribution remains balanced. EQUIVALENCE maps to itself, while FORWARD_ENTAILMENT and BACKWARD_ENTAILMENT swap under the reversal operator:

Gold label	Ordered instances	Percentage
EQUIVALENCE	1,372	35.3
FORWARD_ENTAILMENT	1,260	32.4
BACKWARD_ENTAILMENT	1,260	32.4
Total	3,892	100.0

This distribution reduces the risk that HardCons can be inflated by a trivial majority-class strategy on the reversible subset. In the official release, we will report label distributions for every split and language in `data/README.md`, including the trial and final data.

The `NEGATIVE_OTHER` items introduced from non-reversible PhrasIS labels will be controlled separately. They are useful as noise and for making the classification problem less artificially clean, but they will not dominate the reversible labels and they will not be included in `SoftCons` or `HardCons`, because they do not have a deterministic reversed counterpart. If translation filtering or item removal changes the Spanish or Basque label distribution substantially, we will use stratified sampling or controlled downsampling to keep the official splits comparable across languages, and we will document any such adjustment.

(R2.4) Starter kit specifics. The proposal mentions that a starter kit will be provided, but does not specify which baseline models will be included. Given the hyperparameter sensitivity noted in the pilot study, providing at least one reasonably tuned encoder-only and one decoder-only baseline with documented hyperparameter settings would likely lower the barrier to participation for teams with limited compute resources.

Answer R2.4 We agree that the starter kit should be specified more concretely. The starter kit is intended to provide runnable and reproducible reference systems, not necessarily the strongest systems from the pilot. For that reason, we will prioritize smaller models from each architecture family so that participants with limited GPU resources can execute the baselines.

We will release the official scorer, format validator, data-loading utilities, and baseline training/evaluation scripts. The planned starter-kit baselines are:

Family	Planned baseline	Rationale
Encoder-only	<code>roberta-base</code> or <code>deberta-v3-base</code>	Compact supervised NLI baseline with standard fine-tuning
Decoder-only	<code>OPT-125M</code> or <code>GPT-2 small</code>	Small causal-LM baseline suitable for low-resource GPU settings
Encoder-decoder	<code>BART-base</code> or <code>Flan-T5-base</code>	Smaller sequence-to-sequence baseline covering the encoder-decoder family

For each baseline, we will provide the model identifier, preprocessing steps, train/dev split handling, prediction format, evaluation command, and recommended hyperparameter configuration. The documented configuration will include learning rate, batch size, number of epochs, maximum sequence length, optimizer, random seed policy, and model-selection criterion.

Where redistribution and storage constraints allow it, we will also provide trained checkpoints. If checkpoint redistribution is not practical for a given model, we will

provide the exact model ID, configuration files, and scripts needed to reproduce the baseline. This should make the starter kit useful for participation and sanity checking without requiring participants to tune large models or reproduce the strongest pilot systems.

(R2.5) Scope of the “compositional” claim. The task title includes the term “Compositional,” but the proposal does not explicitly elaborate on the compositional dimension beyond the phrase-level granularity of PhrasIS.

Answer R2.5 We agree that the compositional dimension should be made explicit. In this task, “compositional” is not used to claim a controlled benchmark of arbitrary compositional generalization, where systems are tested on systematically generated syntactic templates. Rather, the term comes from the iSTS/PhrasIS annotation lineage: the original sentence-level STS data was decomposed into aligned semantic segments, or chunks, and these chunks correspond to phrase-level units whose meaning is built from smaller lexical and syntactic material.

The Interpretable STS framework explicitly moves beyond word-level similarity by annotating typed alignments between sentence segments. These segments include noun phrases, verb chains, prepositional phrases, adverbial expressions, and other phrase-level units. The resulting phrase pairs therefore encode how concepts are aggregated into larger local meanings inside a sentence. PhrasIS then reannotates these aligned phrase pairs without sentence context using fine-grained Natural Logic-style labels, making it possible to evaluate whether systems recognize the inferential relation between composed phrase meanings.

We will clarify this in the revised proposal. DiCo-NLI is compositional in this phrase-level sense: it evaluates inference over composed semantic units, not isolated words and not full sentence pairs. However, to avoid overclaiming, the shared task will be framed primarily as **Directional Consistency in Fine-Grained Natural Language Inference**. Analyses by construction type, such as nominal modification, quantification, verb-argument structure, or prepositional attachment, can be included as diagnostic analyses, but the official task target remains fine-grained directional NLI and consistency under reversal.

(R2.6) It would be helpful to clarify whether the task is intended to target specific forms of compositional inference (e.g., adjective-noun or verb-argument compositions), or whether the term is being used more broadly in contrast to sentence-level NLI.

Answer R2.6 The task is not restricted to one specific form of compositional inference, such as adjective-noun composition or verb-argument composition. The term is used more broadly to distinguish phrase-level fine-grained NLI from both word-level lexical relation prediction and full sentence-level NLI.

The official prediction problem is defined over phrase pairs and their reversed counterparts, independently of the internal construction type of the phrase. This

means that all eligible phrase pairs share the same label inventory and reversal operator: `EQUIVALENCE`, `FORWARD_ENTAILMENT`, and `BACKWARD_ENTAILMENT` for the reversible subset, plus `NEGATIVE_OTHER` for non-reversible/noise items.

That said, the construction type is scientifically relevant. PhrasIS includes different phrase-level configurations, including nominal modification, noun-noun expressions, verb chains, prepositional phrases, quantificational expressions, and other chunk types inherited from the iSTS segmentation. We will document these categories where they are available or can be reliably inferred, and we may report diagnostic analyses by phrase type. However, these analyses will be secondary. The shared-task ranking will evaluate the general ability to assign fine-grained directional NLI labels and maintain consistency under reversal across heterogeneous phrase-level semantic units.

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Questions for Authors

1. Translation and adjudication protocol: Will the bilingual annotators for the Spanish and Basque tracks be asked to verify the logical coherence of the reversed pairs in the target language, or only the fidelity of the translation? If a translated phrase pair loses its directional entailment relationship due to morphosyntactic differences (especially in Basque), how will such cases be

handled?

Answer The bilingual annotators will verify both translation fidelity and preservation of the target logical relation. The translation protocol is therefore not limited to checking whether each phrase is fluent or faithful in isolation. Annotators will validate the phrase pair as an NLI instance, including the original direction and the reversed direction under the deterministic label-reversal operator.

For each target language, Spanish and Basque, two bilingual annotators will independently validate a pilot audit of 100 source pairs. This audit will check: (i) phrase-level translation adequacy, (ii) preservation of the expected NLI label, and (iii) preservation of the reversed label after swapping premise and hypothesis. We will report raw agreement, Cohen’s kappa, adjudication counts, and translation-induced label-flip rates.

For the full release, each item will be reviewed by one bilingual annotator, with second-annotator adjudication for low-confidence, ambiguous, or relation-changing cases. If a translated pair loses its intended directional relation because of morphosyntactic differences, scope, case marking, or lexicalization differences, it will be corrected if possible. If the relation cannot be preserved reliably in the target language, the item will be removed from the official evaluation split.

2. Starter kit baselines: Which specific models will be included in the starter kit? Will you provide pre-trained checkpoints, or only training scripts? Given the hyperparameter sensitivity reported in the pilot, will you provide recommended hyperparameter configurations for each architecture family?

Answer The starter kit is intended to provide runnable reference systems, not necessarily the strongest systems from the pilot. We will therefore prioritize smaller models from each architecture family to maximize the chance that participants can run them on modest GPU resources.

The planned starter-kit baselines are:

Family	Planned baseline	Rationale
Encoder-only	roberta-base or deberta-v3-base	Compact supervised NLI baseline with standard fine-tuning
Decoder-only	OPT-125M or GPT-2 small	Small causal-LM baseline suitable for low-resource GPU settings
Encoder-decoder	BART-base or Flan-T5-base	Smaller sequence-to-sequence baseline covering the encoder-decoder family

For each baseline, we will provide the model identifier, preprocessing code, training/evaluation scripts, prediction format, scorer command, and recommended hyperparameter configuration. The documented configuration will include learning rate, batch size, number of epochs, maximum sequence length, optimizer, random seed policy, and model-selection criterion.

Where redistribution and storage constraints allow it, we will also provide trained checkpoints. If checkpoint redistribution is not practical for a given model, we will provide the exact model ID, configuration files, and scripts needed to reproduce the baseline.

3. Label distribution: What is the distribution of EQUIVALENCE, FORWARD-ENTAILMENT, and BACKWARD-ENTAILMENT labels in the pilot dataset? Is it balanced, and if not, how will you address potential class imbalance in the official task data?

Answer The reversible English source subset used in the pilot contains 1,946 source phrase pairs and is close to balanced:

Gold label	Source pairs	Percentage
EQUIVALENCE	686	35.3
FORWARD_ENTAILMENT	624	32.1
BACKWARD_ENTAILMENT	636	32.7
Total	1,946	100.0

After adding the reversed counterpart for each source pair, the distribution remains balanced at the ordered-instance level. EQUIVALENCE maps to itself, while FORWARD_ENTAILMENT and BACKWARD_ENTAILMENT swap under the reversal operator. This reduces the risk that **HardCons** can be inflated by a majority-class strategy on the reversible subset.

We will report label distributions for every official split and language in `data/README.md`. The **NEGATIVE_OTHER** items introduced from non-reversible PhrasIS labels will be controlled separately: they make the task less artificially clean, but they will not dominate the reversible labels and will not be included in **SoftCons** or **HardCons**.

4. Compositional dimension: Could you clarify what “compositional” means in the context of this task? Is the task specifically designed to probe compositional inference (e.g., adjective-noun compositions), or is the term used more broadly?

Answer In this task, “compositional” refers to the phrase-level semantic units inherited from the iSTS/PhrasIS annotation lineage. The original iSTS data decomposes sentence-level STS examples into aligned chunks, including noun phrases, verb chains, prepositional phrases, adverbial expressions, and other

phrase-level units. These chunks aggregate lexical and syntactic material into local semantic units.

PhrasIS reannotates these aligned phrase pairs without sentence context using fine-grained Natural Logic-style labels. DiCo-NLI therefore evaluates inference over composed phrase meanings, rather than isolated words or full sentence pairs.

The task is not restricted to a single construction type, such as adjective-noun or verb-argument composition, and it is not intended as a controlled benchmark of arbitrary compositional generalization. The official ranking evaluates fine-grained directional NLI and consistency under reversal across heterogeneous phrase-level units. Construction-specific analyses, for example by nominal modification, quantification, prepositional attachment, or verb-argument structure, can be included as secondary diagnostics where the categories are available or can be reliably inferred.